Writing Exercises

312. Vic thinks the quotient $\frac{x^{20}}{x^4}$ simplifies to x^5. What is wrong with his reasoning?

313. Mai simplifies the quotient $\frac{y^3}{y}$ by writing $\frac{\cancel{y}^3}{\cancel{y}} = 3$. What is wrong with her reasoning?

314. When Dimple simplified -3^0 and $(-3)^0$ she got the same answer. Explain how using the Order of Operations correctly gives different answers.

315. Roxie thinks n^0 simplifies to 0. What would you say to convince Roxie she is wrong?

Self Check

ⓐ *After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.*

I can...	Confidently	With some help	No-I don't get it!
simplify expressions using the Quotient Property for Exponents.			
simplify expressions with zero exponents.			
simplify expressions using the Quotient to a Power Property.			
simplify expressions by applying several properties.			
divide monomials.			

ⓑ *On a scale of 1–10, how would you rate your mastery of this section in light of your responses on the checklist? How can you improve this?*

10.5 Integer Exponents and Scientific Notation

Learning Objectives

By the end of this section, you will be able to:

- Use the definition of a negative exponent
- Simplify expressions with integer exponents
- Convert from decimal notation to scientific notation
- Convert scientific notation to decimal form
- Multiply and divide using scientific notation

☑ **BE PREPARED** 10.12 Before you get started, take this readiness quiz.

What is the place value of the 6 in the number $64{,}891$?
If you missed this problem, review Example 1.3.

☑ **BE PREPARED** 10.13 Name the decimal 0.0012.
If you missed this problem, review Example 5.1.

☑ **BE PREPARED** 10.14 Subtract: $5-(-3)$.
If you missed this problem, review Example 3.37.

Use the Definition of a Negative Exponent

The Quotient Property of Exponents, introduced in Divide Monomials, had two forms depending on whether the exponent in the numerator or denominator was larger.

Quotient Property of Exponents

If a is a real number, $a \neq 0$, and $m,\ n$ are whole numbers, then

$$\frac{a^m}{a^n} = a^{m-n}, \quad m > n \quad \text{and} \quad \frac{a^m}{a^n} = \frac{1}{a^{n-m}}, \quad n > m$$

What if we just subtract exponents, regardless of which is larger? Let's consider $\frac{x^2}{x^5}$.

We subtract the exponent in the denominator from the exponent in the numerator.

$$\frac{x^2}{x^5}$$
$$x^{2-5}$$
$$x^{-3}$$

We can also simplify $\frac{x^2}{x^5}$ by dividing out common factors: $\frac{x^2}{x^5}$.

$$\frac{\cancel{x} \cdot \cancel{x}}{\cancel{x} \cdot \cancel{x} \cdot x \cdot x \cdot x}$$
$$\frac{1}{x^3}$$

This implies that $x^{-3} = \frac{1}{x^3}$ and it leads us to the definition of a **negative exponent**.

Negative Exponent

If n is a positive integer and $a \neq 0$, then $a^{-n} = \frac{1}{a^n}$.

The negative exponent tells us to re-write the expression by taking the reciprocal of the base and then changing the sign of the exponent. Any expression that has negative exponents is not considered to be in simplest form. We will use the definition of a negative exponent and other properties of exponents to write an expression with only positive exponents.

EXAMPLE 10.63

Simplify:

ⓐ 4^{-2} ⓑ 10^{-3}

Solution

ⓐ

	4^{-2}
Use the definition of a negative exponent, $a^{-n} = \frac{1}{a^n}$.	$\frac{1}{4^2}$
Simplify.	$\frac{1}{16}$

ⓑ

	10^{-3}

Use the definition of a negative exponent, $a^{-n} = \frac{1}{a^n}$.	$\frac{1}{10^3}$
Simplify.	$\frac{1}{1000}$

TRY IT 10.125 Simplify:

ⓐ 2^{-3} ⓑ 10^{-2}

TRY IT 10.126 Simplify:

ⓐ 3^{-2} ⓑ 10^{-4}

When simplifying any expression with exponents, we must be careful to correctly identify the base that is raised to each exponent.

EXAMPLE 10.64

Simplify:

ⓐ $(-3)^{-2}$ ⓑ -3^{-2}

Solution

The negative in the exponent does not affect the sign of the base.

ⓐ

The exponent applies to the base, -3.	$(-3)^{-2}$
Take the reciprocal of the base and change the sign of the exponent.	$\frac{1}{(-3)^2}$
Simplify.	$\frac{1}{9}$

ⓑ

The expression -3^{-2} means "find the opposite of 3^{-2}". The exponent applies only to the base, 3.	-3^{-2}
Rewrite as a product with −1.	$-1 \cdot 3^{-2}$
Take the reciprocal of the base and change the sign of the exponent.	$-1 \cdot \frac{1}{3^2}$
Simplify.	$-\frac{1}{9}$

TRY IT 10.127 Simplify:

ⓐ $(-5)^{-2}$ ⓑ -5^{-2}

TRY IT 10.128 Simplify:

ⓐ $(-2)^{-2}$ ⓑ -2^{-2}

We must be careful to follow the order of operations. In the next example, parts ⓐ and ⓑ look similar, but we get different results.

EXAMPLE 10.65

Simplify:

ⓐ $4 \cdot 2^{-1}$ ⓑ $(4 \cdot 2)^{-1}$

Solution

Remember to always follow the order of operations.

ⓐ	
Do exponents before multiplication.	$4 \cdot 2^{-1}$
Use $a^{-n} = \frac{1}{a^n}$.	$4 \cdot \frac{1}{2^1}$
Simplify.	2

ⓑ	$(4 \cdot 2)^{-1}$
Simplify inside the parentheses first.	$(8)^{-1}$
Use $a^{-n} = \frac{1}{a^n}$.	$\frac{1}{8^1}$
Simplify.	$\frac{1}{8}$

TRY IT 10.129 Simplify:

ⓐ $6 \cdot 3^{-1}$ ⓑ $(6 \cdot 3)^{-1}$

TRY IT 10.130 Simplify:

ⓐ $8 \cdot 2^{-2}$ ⓑ $(8 \cdot 2)^{-2}$

When a variable is raised to a negative exponent, we apply the definition the same way we did with numbers.

EXAMPLE 10.66

Simplify: x^{-6}.

Solution

	x^{-6}
Use the definition of a negative exponent, $a^{-n} = \frac{1}{a^n}$.	$\frac{1}{x^6}$

TRY IT 10.131 Simplify: y^{-7}.

TRY IT 10.132 Simplify: z^{-8}.

When there is a product and an exponent we have to be careful to apply the exponent to the correct quantity. According to the order of operations, expressions in parentheses are simplified before exponents are applied. We'll see how this works in the next example.

EXAMPLE 10.67

Simplify:

ⓐ $5y^{-1}$ ⓑ $(5y)^{-1}$ ⓒ $(-5y)^{-1}$

Solution

ⓐ

Notice the exponent applies to just the base y.	$5y^{-1}$
Take the reciprocal of y and change the sign of the exponent.	$5 \cdot \frac{1}{y^1}$
Simplify.	$\frac{5}{y}$

ⓑ

Here the parentheses make the exponent apply to the base $5y$.	$(5y)^{-1}$
Take the reciprocal of $5y$ and change the sign of the exponent.	$\frac{1}{(5y)^1}$
Simplify.	$\frac{1}{5y}$

ⓒ

	$(-5y)^{-1}$
The base is $-5y$. Take the reciprocal of $-5y$ and change the sign of the exponent.	$\frac{1}{(-5y)^1}$
Simplify.	$\frac{1}{-5y}$
Use $\frac{a}{-b} = -\frac{a}{b}$.	$-\frac{1}{5y}$

TRY IT 10.133 Simplify:

ⓐ $8p^{-1}$ ⓑ $(8p)^{-1}$ ⓒ $(-8p)^{-1}$

TRY IT 10.134 Simplify:

ⓐ $11q^{-1}$ ⓑ $(11q)^{-1}$ ⓒ $(-11q)^{-1}$

Now that we have defined negative exponents, the Quotient Property of Exponents needs only one form, $\frac{a^m}{a^n} = a^{m-n}$, where $a \neq 0$ and m and n are integers.

When the exponent in the denominator is larger than the exponent in the numerator, the exponent of the quotient will

be negative. If the result gives us a negative exponent, we will rewrite it by using the definition of negative exponents, $a^{-n} = \frac{1}{a^n}$.

Simplify Expressions with Integer Exponents

All the exponent properties we developed earlier in this chapter with whole number exponents apply to integer exponents, too. We restate them here for reference.

Summary of Exponent Properties

If a, b are real numbers and m, n are integers, then

Product Property	$a^m \cdot a^n = a^{m+n}$
Power Property	$(a^m)^n = a^{m \cdot n}$
Product to a Power Property	$(ab)^m = a^m b^m$
Quotient Property	$\frac{a^m}{a^n} = a^{m-n},\ a \neq 0$
Zero Exponent Property	$a^0 = 1,\ a \neq 0$
Quotient to a Power Property	$\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m},\ b \neq 0$
Definition of Negative Exponent	$a^{-n} = \frac{1}{a^n}$

EXAMPLE 10.68

Simplify:

ⓐ $x^{-4} \cdot x^6$ ⓑ $y^{-6} \cdot y^4$ ⓒ $z^{-5} \cdot z^{-3}$

Solution

ⓐ

	$x^{-4} \cdot x^6$
Use the Product Property, $a^m \cdot a^n = a^{m+n}$.	x^{-4+6}
Simplify.	x^2

ⓑ

	$y^{-6} \cdot y^4$
The bases are the same, so add the exponents.	y^{-6+4}
Simplify.	y^{-2}
Use the definition of a negative exponent, $a^{-n} = \frac{1}{a^n}$.	$\frac{1}{y^2}$

ⓒ

	$z^{-5} \cdot z^{-3}$

The bases are the same, so add the exponents.	z^{-5-3}
Simplify.	z^{-8}
Use the definition of a negative exponent, $a^{-n}=\frac{1}{a^n}$.	$\frac{1}{z^8}$

TRY IT 10.135 Simplify:

ⓐ $x^{-3}\cdot x^7$ ⓑ $y^{-7}\cdot y^2$ ⓒ $z^{-4}\cdot z^{-5}$

TRY IT 10.136 Simplify:

ⓐ $a^{-1}\cdot a^6$ ⓑ $b^{-8}\cdot b^4$ ⓒ $c^{-8}\cdot c^{-7}$

In the next two examples, we'll start by using the Commutative Property to group the same variables together. This makes it easier to identify the like bases before using the Product Property of Exponents.

EXAMPLE 10.69

Simplify: $(m^4n^{-3})(m^{-5}n^{-2})$.

Solution

	$(m^4n^{-3})(m^{-5}n^{-2})$
Use the Commutative Property to get like bases together.	$m^4m^{-5}\cdot n^{-2}n^{-3}$
Add the exponents for each base.	$m^{-1}\cdot n^{-5}$
Take reciprocals and change the signs of the exponents.	$\frac{1}{m^1}\cdot\frac{1}{n^5}$
Simplify.	$\frac{1}{mn^5}$

TRY IT 10.137 Simplify: $(p^6q^{-2})(p^{-9}q^{-1})$.

TRY IT 10.138 Simplify: $(r^5s^{-3})(r^{-7}s^{-5})$.

If the monomials have numerical coefficients, we multiply the coefficients, just as we did in Use Multiplication Properties of Exponents.

EXAMPLE 10.70

Simplify: $(2x^{-6}y^8)(-5x^5y^{-3})$.

Solution

	$(2x^{-6}y^8)(-5x^5y^{-3})$
Rewrite with the like bases together.	$2(-5)\cdot(x^{-6}x^5)\cdot(y^8y^{-3})$

Simplify.	$-10 \cdot x^{-1} \cdot y^5$
Use the definition of a negative exponent, $a^{-n} = \frac{1}{a^n}$.	$-10 \cdot \frac{1}{x^1} \cdot y^5$
Simplify.	$\frac{-10y^5}{x}$

TRY IT 10.139 Simplify: $(3u^{-5}v^7)(-4u^4v^{-2})$.

TRY IT 10.140 Simplify: $(-6c^{-6}d^4)(-5c^{-2}d^{-1})$.

In the next two examples, we'll use the Power Property and the Product to a Power Property.

EXAMPLE 10.71

Simplify: $\left(k^3\right)^{-2}$.

Solution

	$\left(k^3\right)^{-2}$
Use the Product to a Power Property, $(ab)^m = a^m b^m$.	$k^{3(-2)}$
Simplify.	k^{-6}
Rewrite with a positive exponent.	$\frac{1}{k^6}$

TRY IT 10.141 Simplify: $\left(x^4\right)^{-1}$.

TRY IT 10.142 Simplify: $\left(y^2\right)^{-2}$.

EXAMPLE 10.72

Simplify: $\left(5x^{-3}\right)^2$.

Solution

	$\left(5x^{-3}\right)^2$
Use the Product to a Power Property, $(ab)^m = a^m b^m$.	$5^2\left(x^{-3}\right)^2$
Simplify 5^2 and multiply the exponents of x using the Power Property, $(a^m)^n = a^{m \cdot n}$.	$25x^{-6}$

Rewrite x^{-6} by using the definition of a negative exponent, $a^{-n} = \frac{1}{a^n}$.	$25 \cdot \frac{1}{x^6}$
Simplify	$\frac{25}{x^6}$

TRY IT 10.143 Simplify: $\left(8a^{-4}\right)^2$.

TRY IT 10.144 Simplify: $\left(2c^{-4}\right)^3$.

To simplify a fraction, we use the Quotient Property.

EXAMPLE 10.73

Simplify: $\frac{r^5}{r^{-4}}$.

Solution

	$\frac{r^5}{r^{-4}}$
Use the Quotient Property, $\frac{a^m}{a^n} = a^{m-n}$.	$r^{5-(-4)}$
Be careful to subtract 5 – (–4).	
Simplify.	r^9

TRY IT 10.145 Simplify: $\frac{x^8}{x^{-3}}$.

TRY IT 10.146 Simplify: $\frac{y^7}{y^{-6}}$.

Convert from Decimal Notation to Scientific Notation

Remember working with place value for whole numbers and decimals? Our number system is based on powers of 10. We use tens, hundreds, thousands, and so on. Our decimal numbers are also based on powers of tens—tenths, hundredths, thousandths, and so on.

Consider the numbers 4000 and 0.004. We know that 4000 means 4×1000 and 0.004 means $4 \times \frac{1}{1000}$. If we write the 1000 as a power of ten in exponential form, we can rewrite these numbers in this way:

4000	0.004
4×1000	$4 \times \frac{1}{1000}$
4×10^3	$4 \times \frac{1}{10^3}$
	4×10^{-3}

When a number is written as a product of two numbers, where the first factor is a number greater than or equal to one but less than 10, and the second factor is a power of 10 written in exponential form, it is said to be in *scientific notation.*

Scientific Notation

A number is expressed in **scientific notation** when it is of the form

$$a \times 10^n$$

where $a \geq 1$ and $a < 10$ and n is an integer.

It is customary in scientific notation to use $\times$ as the multiplication sign, even though we avoid using this sign elsewhere in algebra.

Scientific notation is a useful way of writing very large or very small numbers. It is used often in the sciences to make calculations easier.

If we look at what happened to the decimal point, we can see a method to easily convert from decimal notation to scientific notation.

$4000. = 4 \times 10^3$	$0.004 = 4 \times 10^{-3}$
$4000. = 4 \times 10^3$	$0.004 = 4 \times 10^{-3}$
Moved the decimal point 3 places to the left.	Moved the decimal point 3 places to the right.

In both cases, the decimal was moved 3 places to get the first factor, 4, by itself.

- The power of 10 is positive when the number is larger than 1: $4000 = 4 \times 10^3$.
- The power of 10 is negative when the number is between 0 and 1: $0.004 = 4 \times 10^{-3}$.

EXAMPLE 10.74

Write $37{,}000$ in scientific notation.

Solution

Step 1: Move the decimal point so that the first factor is greater than or equal to 1 but less than 10.	$37000.$
Step 2: Count the number of decimal places, n, that the decimal point was moved.	3.70000 4 places
Step 3: Write the number as a product with a power of 10.	3.7×10^4
If the original number is: • greater than 1, the power of 10 will be 10^n. • between 0 and 1, the power of 10 will be 10^{-n}	
Step 4: Check.	
10^4 is 10,000 and 10,000 times 3.7 will be 37,000.	
	$37{,}000 = 3.7 \times 10^4$

TRY IT 10.147 Write in scientific notation: $96{,}000$.

TRY IT 10.148 Write in scientific notation: $48{,}300$.

HOW TO

Convert from decimal notation to scientific notation.

Step 1. Move the decimal point so that the first factor is greater than or equal to 1 but less than 10.
Step 2. Count the number of decimal places, n, that the decimal point was moved.
Step 3. Write the number as a product with a power of 10.
- If the original number is:
  - greater than 1, the power of 10 will be 10^n.
  - between 0 and 1, the power of 10 will be 10^{-n}.

Step 4. Check.

EXAMPLE 10.75

Write in scientific notation: 0.0052.

Solution

	0.0052
Move the decimal point to get 5.2, a number between 1 and 10.	0.0052
Count the number of decimal places the point was moved.	3 places
Write as a product with a power of 10.	5.2×10^{-3}
Check your answer: 5.2×10^{-3} $5.2 \times \frac{1}{10^3}$ $5.2 \times \frac{1}{1000}$ 5.2×0.001 0.0052	
	$0.0052 = 5.2 \times 10^{-3}$

TRY IT 10.149 Write in scientific notation: 0.0078.

TRY IT 10.150 Write in scientific notation: 0.0129.

Convert Scientific Notation to Decimal Form

How can we convert from scientific notation to decimal form? Let's look at two numbers written in scientific notation and see.

9.12×10^4	9.12×10^{-4}
$9.12 \times 10{,}000$	9.12×0.0001
$91{,}200$	0.000912

If we look at the location of the decimal point, we can see an easy method to convert a number from scientific notation to decimal form.

$9.12 \times 10^{4} = 91{,}200$ $\quad$ $9.12 \times 10^{-4} = 0.000912$

$9.12__ \times 10^{4} = 91{,}200$ $\quad$ $___9.12 \times 10^{-4} = 0.000912$

In both cases the decimal point moved 4 places. When the exponent was positive, the decimal moved to the right. When the exponent was negative, the decimal point moved to the left.

EXAMPLE 10.76

Convert to decimal form: 6.2×10^{3}.

Solution

Step 1: Determine the exponent, n, on the factor 10.	6.2×10^{3}
Step 2: Move the decimal point n places, adding zeros if needed. • If the exponent is positive, move the decimal point n places to the right. • If the exponent is negative, move the decimal point $\lvert n \rvert$ places to the left.	6.200, 6,200
Step 3: Check to see if your answer makes sense.	
10^{3} is 1000 and 1000 times 6.2 will be 6,200.	$6.2 \times 10^{3} = 6{,}200$

TRY IT 10.151 Convert to decimal form: 1.3×10^{3}.

TRY IT 10.152 Convert to decimal form: 9.25×10^{4}.

HOW TO

Convert scientific notation to decimal form.

Step 1. Determine the exponent, n, on the factor 10.
Step 2. Move the decimal n places, adding zeros if needed.
- If the exponent is positive, move the decimal point n places to the right.
- If the exponent is negative, move the decimal point $\lvert n \rvert$ places to the left.

Step 3. Check.

EXAMPLE 10.77

Convert to decimal form: 8.9×10^{-2}.

Solution

	8.9×10^{-2}
Determine the exponent n, on the factor 10.	The exponent is −2.

Move the decimal point 2 places to the left.	-8.9
Add zeros as needed for placeholders.	0.089
	$8.9 \times 10^{-2} = 0.089$
The Check is left to you.	

TRY IT 10.153 Convert to decimal form: 1.2×10^{-4}.

TRY IT 10.154 Convert to decimal form: 7.5×10^{-2}.

Multiply and Divide Using Scientific Notation

We use the Properties of Exponents to multiply and divide numbers in scientific notation.

EXAMPLE 10.78

Multiply. Write answers in decimal form: $(4 \times 10^5)(2 \times 10^{-7})$.

Solution

	$(4 \times 10^5)(2 \times 10^{-7})$
Use the Commutative Property to rearrange the factors.	$4 \cdot 2 \cdot 10^5 \cdot 10^{-7}$
Multiply 4 by 2 and use the Product Property to multiply 10^5 by 10^{-7}.	8×10^{-2}
Change to decimal form by moving the decimal two places left.	0.08

TRY IT 10.155 Multiply. Write answers in decimal form: $(3 \times 10^6)(2 \times 10^{-8})$.

TRY IT 10.156 Multiply. Write answers in decimal form: $(3 \times 10^{-2})(3 \times 10^{-1})$.

EXAMPLE 10.79

Divide. Write answers in decimal form: $\frac{9 \times 10^3}{3 \times 10^{-2}}$.

Solution

	$\frac{9 \times 10^3}{3 \times 10^{-2}}$
Separate the factors.	$\frac{9}{3} \times \frac{10^3}{10^{-2}}$

Divide 9 by 3 and use the Quotient Property to divide 10^3 by 10^{-2}.	3×10^5
Change to decimal form by moving the decimal five places right.	300,000

TRY IT 10.157 Divide. Write answers in decimal form: $\frac{8 \times 10^4}{2 \times 10^{-1}}$.

TRY IT 10.158 Divide. Write answers in decimal form: $\frac{8 \times 10^2}{4 \times 10^{-2}}$.

MEDIA

ACCESS ADDITIONAL ONLINE RESOURCES

Negative Exponents (https://www.openstax.org/l/24negexponents)

Examples of Simplifying Expressions with Negative Exponents (https://www.openstax.org/l/24simpexprnegex)

Scientific Notation (https://www.openstax.org/l/24scnotation)

SECTION 10.5 EXERCISES

Practice Makes Perfect

Use the Definition of a Negative Exponent

In the following exercises, simplify.

316. 5^{-3}

317. 8^{-2}

318. 3^{-4}

319. 2^{-5}

320. 7^{-1}

321. 10^{-1}

322. $2^{-3} + 2^{-2}$

323. $3^{-2} + 3^{-1}$

324. $3^{-1} + 4^{-1}$

325. $10^{-1} + 2^{-1}$

326. $10^0 - 10^{-1} + 10^{-2}$

327. $2^0 - 2^{-1} + 2^{-2}$

328. (a) $(-6)^{-2}$ (b) -6^{-2}

329. (a) $(-8)^{-2}$ (b) -8^{-2}

330. (a) $(-10)^{-4}$ (b) -10^{-4}

331. (a) $(-4)^{-6}$ (b) -4^{-6}

332. (a) $5 \cdot 2^{-1}$ (b) $(5 \cdot 2)^{-1}$

333. (a) $10 \cdot 3^{-1}$ (b) $(10 \cdot 3)^{-1}$

334. (a) $4 \cdot 10^{-3}$ (b) $(4 \cdot 10)^{-3}$

335. (a) $3 \cdot 5^{-2}$ (b) $(3 \cdot 5)^{-2}$

336. n^{-4}

337. p^{-3}

338. c^{-10}

339. m^{-5}

340. (a) $4x^{-1}$ (b) $(4x)^{-1}$ (c) $(-4x)^{-1}$

341. (a) $3q^{-1}$ (b) $(3q)^{-1}$ (c) $(-3q)^{-1}$

342. (a) $6m^{-1}$ (b) $(6m)^{-1}$ (c) $(-6m)^{-1}$

343. (a) $10k^{-1}$ (b) $(10k)^{-1}$ (c) $(-10k)^{-1}$

Simplify Expressions with Integer Exponents

In the following exercises, simplify.

344. $p^{-4} \cdot p^8$

345. $r^{-2} \cdot r^5$

346. $n^{-10} \cdot n^2$

347. $q^{-8} \cdot q^3$

348. $k^{-3} \cdot k^{-2}$

349. $z^{-6} \cdot z^{-2}$

350. $a \cdot a^{-4}$

351. $m \cdot m^{-2}$

352. $p^5 \cdot p^{-2} \cdot p^{-4}$

353. $x^4 \cdot x^{-2} \cdot x^{-3}$

354. $a^3 b^{-3}$

355. $u^2 v^{-2}$

356. $\left(x^5 y^{-1}\right)\left(x^{-10} y^{-3}\right)$

357. $\left(a^3 b^{-3}\right)\left(a^{-5} b^{-1}\right)$

358. $\left(uv^{-2}\right)\left(u^{-5} v^{-4}\right)$

359. $\left(pq^{-4}\right)\left(p^{-6} q^{-3}\right)$

360. $\left(-2r^{-3} s^9\right)\left(6r^4 s^{-5}\right)$

361. $\left(-3p^{-5} q^8\right)\left(7p^2 q^{-3}\right)$

362. $\left(-6m^{-8} n^{-5}\right)\left(-9m^4 n^2\right)$

363. $\left(-8a^{-5} b^{-4}\right)\left(-4a^2 b^3\right)$

364. $\left(a^3\right)^{-3}$

365. $\left(q^{10}\right)^{-10}$

366. $\left(n^2\right)^{-1}$

367. $\left(x^4\right)^{-1}$

368. $\left(y^{-5}\right)^4$

369. $\left(p^{-3}\right)^2$

370. $\left(q^{-5}\right)^{-2}$

371. $\left(m^{-2}\right)^{-3}$

372. $\left(4y^{-3}\right)^2$

373. $\left(3q^{-5}\right)^2$

374. $\left(10p^{-2}\right)^{-5}$

375. $\left(2n^{-3}\right)^{-6}$

376. $\frac{u^9}{u^{-2}}$

377. $\frac{b^5}{b^{-3}}$

378. $\frac{x^{-6}}{x^4}$

379. $\frac{m^5}{m^{-2}}$

380. $\frac{q^3}{q^{12}}$

381. $\frac{r^6}{r^9}$

382. $\frac{n^{-4}}{n^{-10}}$

383. $\frac{p^{-3}}{p^{-6}}$

Convert from Decimal Notation to Scientific Notation

In the following exercises, write each number in scientific notation.

384. 45,000

385. 280,000

386. 8,750,000

387. 1,290,000

388. 0.036

389. 0.041

390. 0.00000924

391. 0.0000103

392. The population of the United States on July 4, 2010 was almost 310,000,000.

393. The population of the world on July 4, 2010 was more than 6,850,000,000.

394. The average width of a human hair is 0.0018 centimeters.

395. The probability of winning the 2010 Megamillions lottery is about 0.0000000057.

Convert Scientific Notation to Decimal Form

In the following exercises, convert each number to decimal form.

396. 4.1×10^2

397. 8.3×10^2

398. 5.5×10^8

399. 1.6×10^{10}

400. 3.5×10^{-2}

401. 2.8×10^{-2}

402. 1.93×10^{-5}

403. 6.15×10^{-8}

404. In 2010, the number of Facebook users each day who changed their status to 'engaged' was 2×10^4.

405. At the start of 2012, the US federal budget had a deficit of more than $\$1.5 \times 10^{13}$.

406. The concentration of carbon dioxide in the atmosphere is 3.9×10^{-4}.

407. The width of a proton is 1×10^{-5} of the width of an atom.